

A Unified SME $\leftrightarrow$ Exotic-Potential Framework for CPT Symmetry Tests

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Research Question

Light bosons predicted in many extensions of the Standard Model (axion-like particles, dark-sector mediators, Lorentz- and CPT-violating fields) produce spin-dependent forces between fermions. Hundreds of published bounds, from NV-centre magnetometry, torsion balances, co-magnetometers and precision spectroscopy, constrain their couplings. Matter-sector and antimatter-sector results had not been compared systematically, although that comparison is a natural test of CPT symmetry. Can it be done with the data that exist?

Framework

The Standard Model Extension (SME) describes Lorentz and CPT violation through fixed background coefficients in the fermion Lagrangian. A canonical Foldy–Wouthuysen reduction maps these onto the non-relativistic Dobrescu–Mocioiu potentials V_1 – V_{16} , the basis in which experiments report their limits. SPINDEP, the pipeline I built, compiles the published constraint curves, matches each matter dataset to its antimatter counterpart (for example e – e with e – e^+), interpolates both onto a common interaction-range grid λ , and evaluates

$$A_\alpha(\lambda) = \frac{g_\alpha^f(\lambda) - g_\alpha^{\bar{f}}(\lambda)}{g_\alpha^f(\lambda) + g_\alpha^{\bar{f}}(\lambda)}.$$

Status and Current Results

- **Complete spin-dependent dictionary.** b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ and $H_{\mu\nu}$ are the only minimal-SME fermion coefficients with spin-dependent terms through third order in $1/m$; a_μ , $c_{\mu\nu}$ and e_μ give spin-independent terms only, and f_μ is absent. All four reductions agree with Kostelecký and Lane term by term.
- **Matter versus antimatter.** Between CPT-conjugate states, the CPT-odd b_μ and $g_{\lambda\mu\nu}$ change sign and the CPT-even $d_{\mu\nu}$ and $H_{\mu\nu}$ do not. Checked symbolically for all 16 coefficient families.
- **Database.** 283 constraint datasets compiled, 247 analysed; only 22 involve an antimatter sector (antiprotonic helium, positronium, muonium, $dd\mu^+$).
- **Matched pairs.** 15 matter–antimatter pairs (13 independent). Mean $|A_\alpha|$ ranges from 0.33 to 1.00, and every pair is significant after correcting for correlated grid points ($Z = 5.1$ to 51).
- **What that significance means.** For two one-sided upper limits $U_f = s_f g_\star$ and $U_{\bar{f}} = s_{\bar{f}} g_\star$,

$$A_U = \frac{U_f - U_{\bar{f}}}{U_f + U_{\bar{f}}} = \frac{s_f - s_{\bar{f}}}{s_f + s_{\bar{f}}},$$

so the coupling cancels and only the sensitivity ratio is left. The data agree: each of the four pairs built on the 2025 hydrogen bounds has a twin with an older matter-side bound, and in every case the tighter matter bound gave the larger asymmetry. The asymmetry measures a sensitivity gap, not CPT violation.

- **Where the gap is.** 7 of 12 potentials and 6 of 11 fermion-sector pairs have no antimatter data. The e – N , n – N and p – N sectors hold 117 matter-sector datasets and none for antimatter.

What I Would Value Feedback On

- **Experimental grounding:** are the dataset pairings faithful to the experiments, and are there antimatter results I have missed, especially in the nucleon sectors?
- **Signed observables:** which antimatter platforms (BASE, ALPHA, positronium, antiprotonic helium) could report signed shifts or likelihoods, so that a genuine CPT test replaces the comparison of upper limits?
- **Two-body matching:** the dictionary is exact at the level of one-body operators. What is the best route to matching explicit mediator amplitudes?